IndicDoc Verify

- **Project title:** IndicDoc Verify: Deep-Learning Layout Analysis for Pre-Print Checking of Indian Question Papers and Their Translations
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- **Course:** DL Mini Project (Expt. 11, SDG-based)

- Department of Artificial Intelligence and Machine Learning
- St. Francis Institute of Technology (SFIT)

- **Project guide:** Dr. Joanne Gomes
- **Academic year / semester:** 2026–27, Semester V (TE AIML)

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Literature Survey Summary Table

Ref	Author	Problem	Dataset	Method	Limitation
[1]	Nath et al., 2025	Layout parsing of Indian-language pages	IndicDLP: 119,806 pages, 12 languages, 42 labels incl. questions and options	Benchmark of YOLOv10x, DocLayout-YOLO, RoDLA, DINO, DiT, Florence-2	Detects regions only; nobody uses the question labels to check papers
[2]	Kasuba et al., 2025	Page-level handwritten Indic text recognition	CHIPS, ~25K pages, 10 languages	DBNet word detection + 6 recognisers (CRNN, PARSeq, ...)	User must name the language; text only
[3]	Mondal et al., 2026	Multilingual table detection	UniTabBank: 55K pages, 82K tables, 28 languages	Semi-automatic labelling; YOLOv11 detector	Borderless tables and degraded scans fail
[4]	Pal et al., 2024	Table question answering, low-resource Indic	BanglaTabQA 2M, HindiTabQA 643K	SQL-generated QA; fine-tuned mBART / M2M / Llama	Wikipedia tables only
[5]	Xu et al., 2021	Multilingual document understanding	30M docs; XFUND, 7 languages	LayoutXLM: text + layout + image transformer	No Indian language
[6]	Kasuba et al., 2025	Grounding answers on the page	MGVG: 70 circulars, 509 QA pairs	OCR + Llama-3.1-8B + region matching	OCR-dependent; single page
[7]	Bhattacharyya et al., 2025	Hindi OCR with vision-language models	1.1M printed Hindi words + pages	LoRA-tuned GOT-OCR 2.0	Hindi only; mixed-language lines are hard
[8]	Dumada & Shah, 2026	Devanagari table extraction	500+ Devanagari documents	Structure + Tesseract content, weighted fusion	No ablation; self-reported
[9]	Babayeva et al., 2025	Translating a test into other languages with an LLM	Force Concept Inventory, many languages	GPT-4o translation + back-translation + human review	Every item still checked by hand; no page structure
[10]	Rei et al., 2022	Translation quality estimation without a reference	WMT 2022 QE shared task	CometKiwi: multilingual encoder predicts quality per sentence	Plain sentences: no question numbers, options or marks

Research Gaps

#	Gap in existing work	How this project addresses it
G1	Exam papers are translated into up to 13 languages and vetted only by hand; errors reach candidates	An automatic pre-print check of every language version against the source
G2	Translation-quality tools [9, 10] read plain sentences: blind to question numbers, options, marks, figures	Compare versions question by question using the page layout itself
G3	Indic layout models [1] detect questions and options, but this is not used to check papers	Use the detector's question / option regions (AP@50 0.81–0.90) as the backbone of the checker
G4	Indic OCR misreads Latin digits in mixed text [7]; a checker of numbers needs every digit right	A "number pass": two OCR models must agree on every number

- Court e-filing checks already exist (Supreme Court + IIT Madras), so we chose what **nobody automates yet**: checking a question paper and its translations.

Problem Statement and Objectives

- **Problem:** exam papers are translated into Hindi and regional languages and checked only by hand; errors reach lakhs of candidates.
- **Main objective:** an offline DL system that flags errors in a paper and every difference from its source version, before printing.
- **Specific objectives:**
  - Detect questions, options, answers with a layout detector (compare 3).
  - Read text with reliable numbers in 12 languages.
  - Check numbering, duplicates, options, marks, parity, repeats.
  - Measure on a benchmark; ship app + API in Docker.
- **Expected output:** verdict + each finding with question and page (HTML / JSON).

It has already happened

Exam	What went wrong
NEET-UG 2018	49 questions wrong in the Tamil version; HC ordered 196 grace marks, SC set it aside
KPSC KAS 2024	59 errors in the Kannada paper; re-exam for ~2 lakh candidates, errors again
APPSC Group-I	51 of 120 questions alleged mistranslated into Telugu; SC notice
UGC-NET Jun 2026	Translation errors + questions repeated from old papers; re-test 9–10 Sep
MBBS (NTRUHS) 2026	The same MCQ printed twice; AP HC ordered a compensatory mark

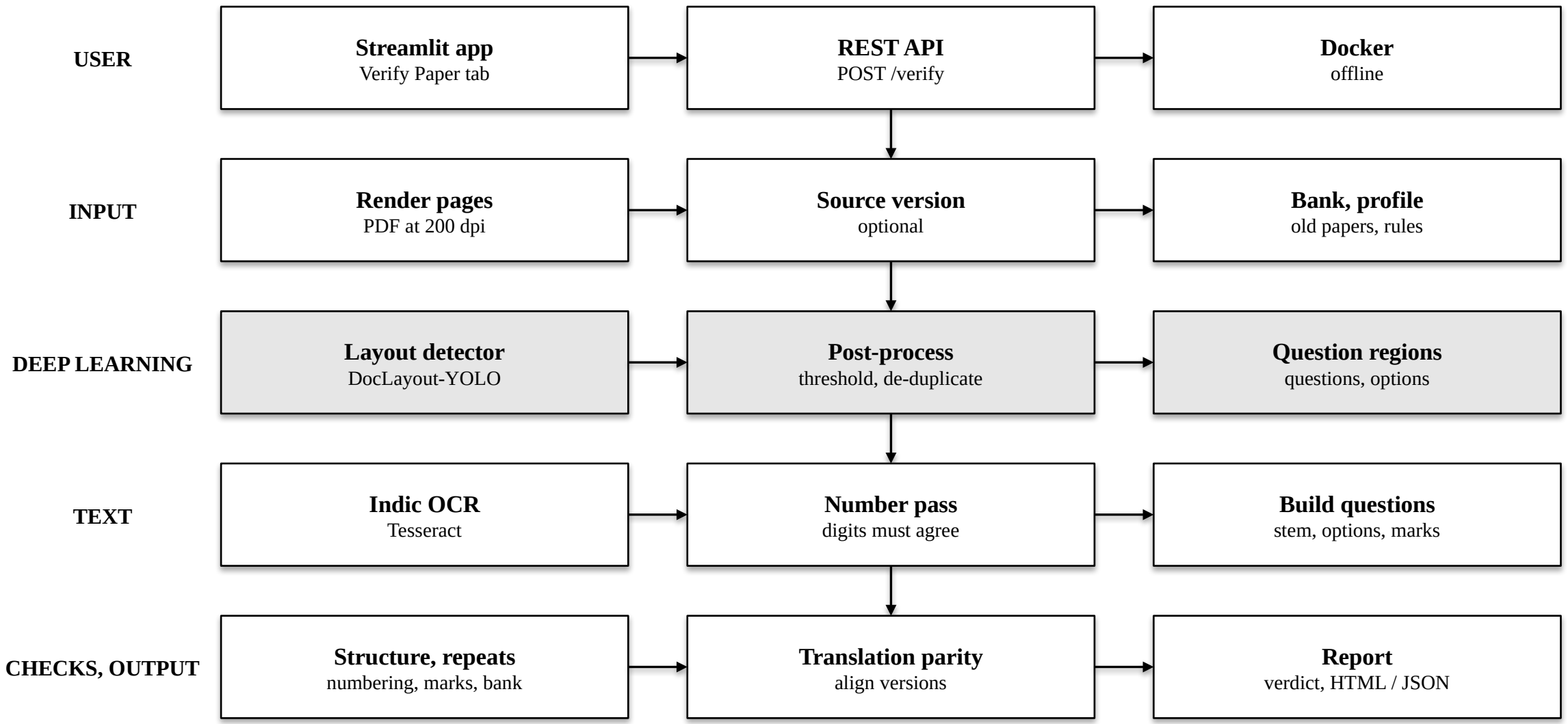
Proposed System, Methodology and Novelty

- **Workflow (input to output):**

1. **Paper in:** PDF / scans + source version
2. **Find structure:** DocLayout-YOLO: questions, options
3. **Read text:** Indic OCR + English number pass
4. **Build questions:** number, stem, options, marks, figures
5. **Check:** structure, parity, repeats, format
6. **Report:** verdict + findings on the page

- **Deep learning:** DocLayout-YOLO finds every question, option and answer (AP@50 0.81–0.90).
- **Novelty 1 – parity without translation:** numbers, options, marks and figures must match in every language.
- **Novelty 2 – number pass:** two OCR models must agree on every digit.
- **USP:** the only tool we found that checks a paper against its translations, offline, in 12 languages.

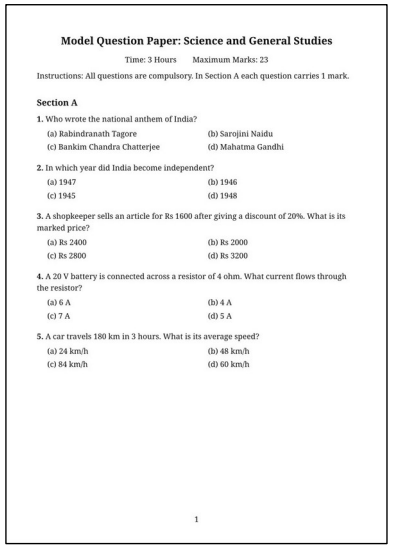
System Block Diagram



Grey: deep-learning stage. The source version follows the same path.

Dataset Description and Pre-processing

- **Dataset and source:** IndicDLP (ICDAR 2025), Hugging Face / AIKosh; 12 languages, 42 labels.
- **Samples:** 3,317 pages, 49,995 boxes, 42 classes.
- **Split:** train 2,293 / val 342 / test 682 pages.
- **Benchmark:** 132 English–Hindi/Tamil paper pairs, 108 with one injected error.
- **Pre-processing:** COCO → YOLO labels, 640 / 1024 px; PDFs at 200 dpi.
- **Augmentation:** mosaic, HSV, scale (YOLO); jitter, blur (R-CNN).
- **Challenges:** class imbalance (paragraph ≈ 32%); no public data of real errors.



Model Question Paper: Science and General Studies

Time: 3 Hours Maximum Marks: 23

Instructions: All questions are compulsory. In Section A each question carries 1 mark.

Section A

1. Who wrote the national anthem of India?
(a) Rabindranath Tagore (b) Sarojini Naidu
(c) Bankim Chandra Chatterjee (d) Mahatma Gandhi

2. In which year did India become independent?
(a) 1947 (b) 1946
(c) 1945 (d) 1948

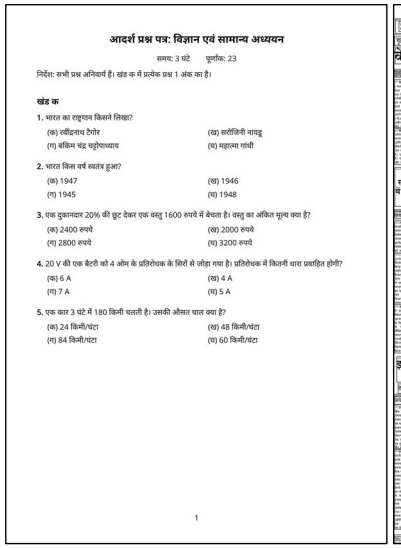
3. A shopkeeper sells an article for Rs 1600 after giving a discount of 20%. What is its marked price?
(a) Rs 2400 (b) Rs 2000
(c) Rs 2800 (d) Rs 3200

4. A 20 V battery is connected across a resistor of 4 ohm. What current flows through the resistor?
(a) 6 A (b) 4 A
(c) 7 A (d) 5 A

5. A car travels 180 km in 3 hours. What is its average speed?
(a) 24 km/h (b) 48 km/h
(c) 84 km/h (d) 60 km/h

1

English source



आदर्श प्रश्न पत्र: विज्ञान एवं सामान्य अध्ययन

समय: 3 घंटे पूर्णांक: 23

निर्देश: सभी प्रश्न अनिवार्य हैं। खंड क में प्रत्येक प्रश्न 1 अंक का है।

खंड क

1. भारत का राष्ट्रगान किसने लिखा?
(क) रवींद्रनाथ टैगोर (ख) सरोजिनी नायडू
(ग) बंकिम चंद्र चट्टोपाध्याय (घ) महात्मा गांधी

2. भारत किस वर्ष स्वतंत्र हुआ?
(क) 1947 (ख) 1946
(ग) 1945 (घ) 1948

3. एक दुकानदार 20% की छूट देकर एक वस्तु 1600 रुपये में बेचता है। वस्तु का अंकित मूल्य क्या है?
(क) 2400 रुपये (ख) 2000 रुपये
(ग) 2800 रुपये (घ) 3200 रुपये

4. 20 V की एक बैटरी को 4 ओम के प्रतिरोधक के सिरों से जोड़ा गया है। प्रतिरोधक में कितनी धारा प्रवाहित होगी?
(क) 6 A (ख) 4 A
(ग) 7 A (घ) 5 A

5. एक कार 3 घंटे में 180 किमी चलती है। उसकी औसत चाल क्या है?
(क) 24 किमी/घंटा (ख) 48 किमी/घंटा
(ग) 84 किमी/घंटा (घ) 60 किमी/घंटा

1

Hindi version



8 | जिलावार | 11 अक्टूबर, 2023 | दैनिक

केसीआर ने गरीबों के लिए काम नहीं किया

'आप' विधायक अमानतुल्लाह के परिसरों पर ईडी के छापे

विशेषाधिकार समिति को भेजकर मामले को खत्म कर सकते हैं स्पीकर

आयुष्मान के भारत को वृद्धि दर का अनुमान बढ़ाकर 6.3 फीसद किया

मस्जिद को हमास से पहले दिया समर्थन से मुकर सकता है मुस्लिम संगठन

माफी की मांग पर निर्णय लेने में कितना समय लगता है?

राजस्थान में महिला निशानेबाजों से बलात्कार व छेड़छाड़, प्रशिक्षक के खिलाफ प्राथमिकी

संजय सिंह की ईडी हिरासत 13 अक्तूबर तक बढ़ी

इजराइल ने गाजा में हमले तेज किए, तीन लाख अतिरिक्त सैनिक तैनात

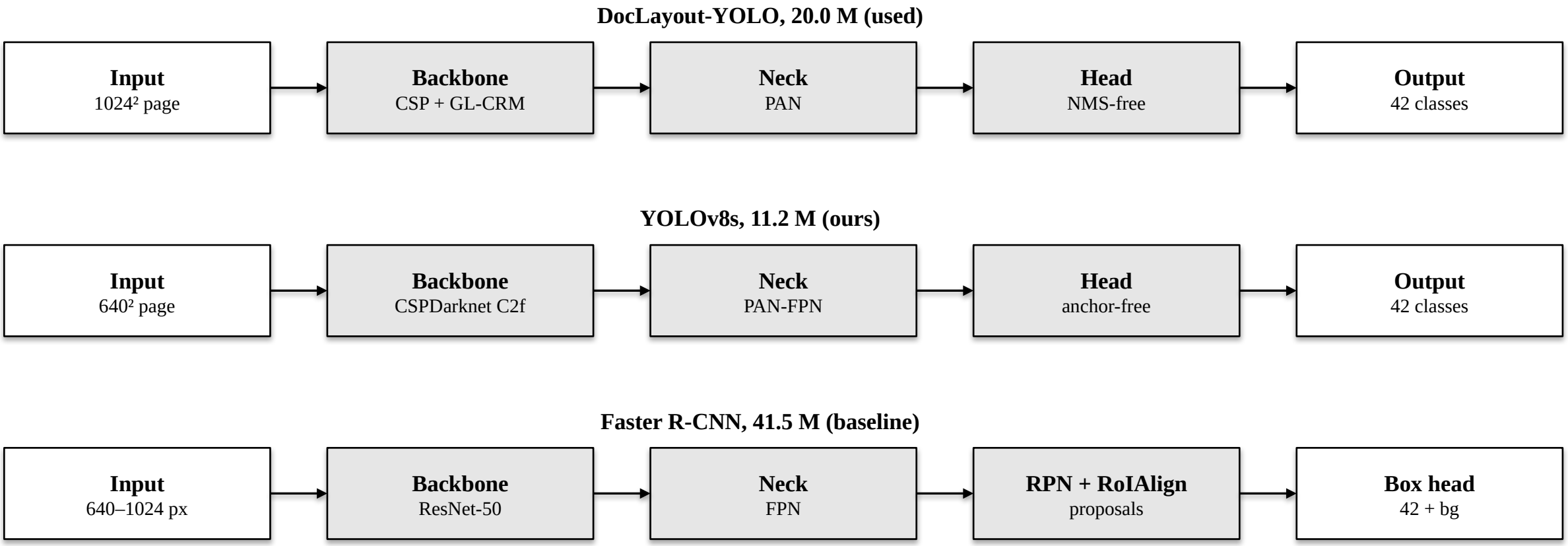
जीवित जलाकर मारने की कोशिश करने वाले पति को उम्रकैद

IndicDLP page

Actual data used

Deep Learning Model Architecture

- **Selected:** DocLayout-YOLO, 20.0 M params, pretrained on IndicDLP by its authors. **Input** page image → **output** boxes with 42 labels. **Modified:** 42-class heads.



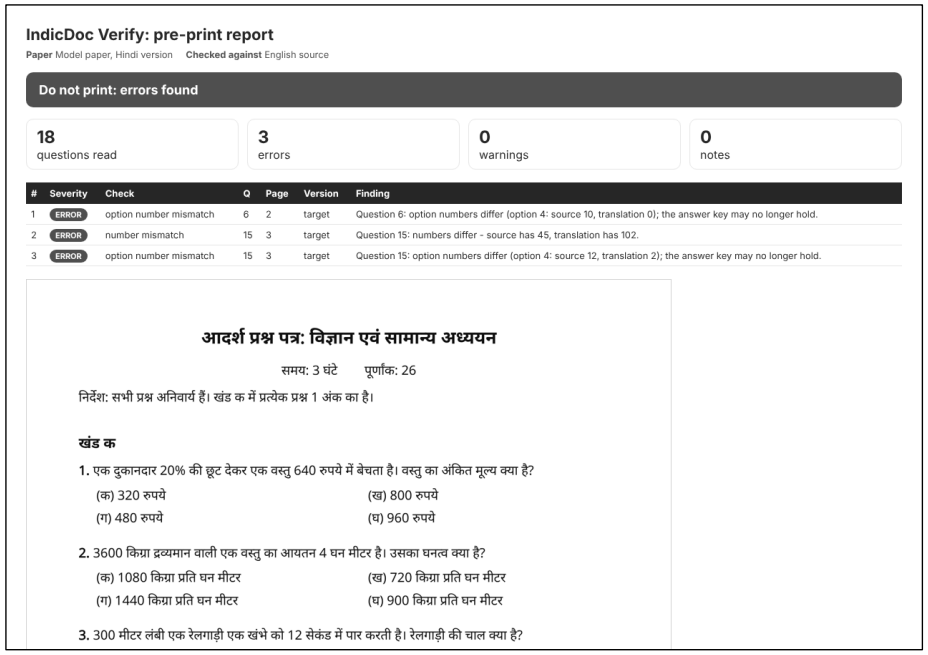
Tools, Technologies and Environment

Component	Details to report
Programming language	Python 3.11 (ML, checks, app, API); JavaScript (Node) for the benchmark renderer
DL framework	PyTorch 2 with torchvision; Ultralytics 8 (YOLOv8); doclayout-yolo 0.0.4
Development environment	VS Code / terminal, Git + GitHub, pytest (unit tests for every check)
Libraries	pycocotools, OpenCV, Pillow, NumPy, pandas, Matplotlib, PyMuPDF; Tesseract 5 (tessdata_best, 12 scripts); tesseract.js for the benchmark
Hardware	Apple M1 Pro GPU via PyTorch MPS (training, evaluation); CPU container for the benchmark; CPU in Docker
Dataset tools	Hugging Face Hub + fsspec (partial Parquet reads); COCO / YOLO converters; Playwright + Noto fonts to render benchmark papers
Interface / deployment	Streamlit app (Verify Paper tab) + FastAPI (/verify); Docker + docker-compose, offline

Implementation Details

- **Built:** checker (ml/verify), OCR + number pass, 2 trained detectors, benchmark, app, API.
- **Data loading:** IndicDLP Parquet → COCO; PDFs at 200 dpi.
- **Model:** pretrained DocLayout-YOLO; YOLOv8s, Faster R-CNN fine-tuned.
- **Loss / optimiser:** CIoU + DFL + BCE; SGD lr 0.005, batch 4.
- **Inference:** layout → OCR → questions → checks → report.
- **Interface:** Streamlit tab + POST /verify.

```
out = predict_single(model, page, "doclayout")
doc = extract_document(page, out, "hin",
                       number_pass=True)
report = verify(hindi_pages,
                source_pages=english_pages)
```



IndicDoc Verify: pre-print report

Paper Model paper, Hindi version **Checked against** English source

Do not print: errors found

18	3	0	0
questions read	errors	warnings	notes

#	Severity	Check	Q	Page	Version	Finding
1	ERROR	option number mismatch	6	2	target	Question 6: option numbers differ (option 4: source 10, translation 0); the answer key may no longer hold.
2	ERROR	number mismatch	15	3	target	Question 15: numbers differ - source has 45, translation has 102.
3	ERROR	option number mismatch	15	3	target	Question 15: option numbers differ (option 4: source 12, translation 2); the answer key may no longer hold.

आदर्श प्रश्न पत्र: विज्ञान एवं सामान्य अध्ययन

समय: 3 घंटे पूर्णांक: 26

निर्देश: सभी प्रश्न अनिवार्य हैं। खंड क में प्रत्येक प्रश्न 1 अंक का है।

खंड क

1. एक दुकानदार 20% की छूट देकर एक वस्तु 640 रुपये में बेचता है। वस्तु का अंकित मूल्य क्या है?
   (क) 320 रुपये (ख) 800 रुपये
   (ग) 480 रुपये (घ) 960 रुपये

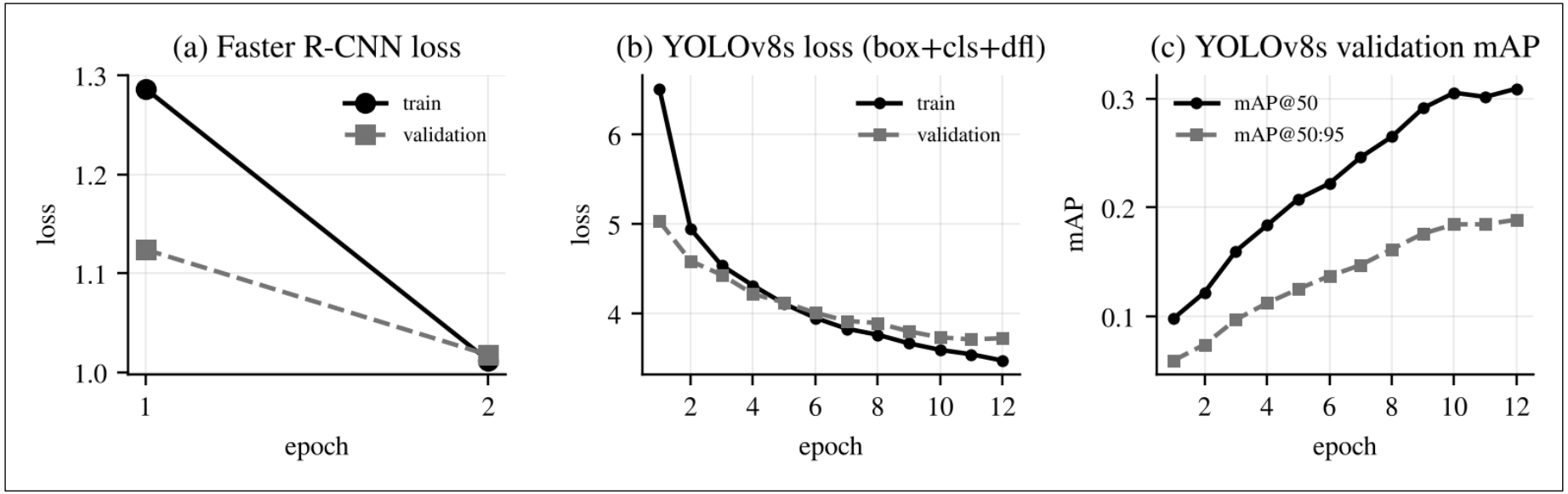
2. 3600 किग्रा द्रव्यमान वाली एक वस्तु का आयतन 4 घन मीटर है। उसका घनत्व क्या है?
   (क) 1080 किग्रा प्रति घन मीटर (ख) 720 किग्रा प्रति घन मीटर
   (ग) 1440 किग्रा प्रति घन मीटर (घ) 900 किग्रा प्रति घन मीटर

3. 300 मीटर लंबी एक रेलगाड़ी एक खंभे को 12 सेकंड में पार करती है। रेलगाड़ी की चाल क्या है?

Report produced by the system

Experimental Setup and Training

- **Data:** train 2,293 / val 342 / test 682; benchmark 132 + 48 dev pairs.
- **Faster R-CNN:** batch 4, SGD 0.005, 2 epochs, 76 min. **YOLOv8s:** 640 px, batch 8, 12 epochs, 60 min (M1 Pro GPU).
- **Experiments:** layout mAP; robustness to scan damage; verification benchmark; ablations.

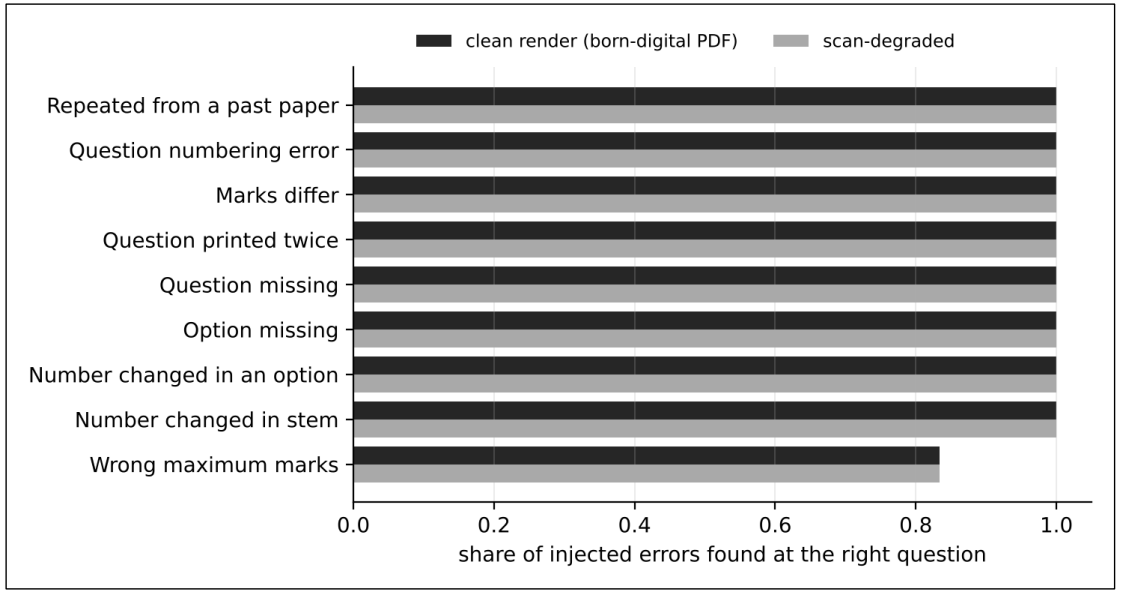


Training / validation curves (measured)

Results and Performance Evaluation

Model	mAP@50	mAP@50:95	Precision	Recall	F1
DocLayout-YOLO	0.759	0.575	0.848	0.775	0.810
YOLOv8s (ours)	0.259	0.165	0.603	0.508	0.551
Faster R-CNN (ours)	0.148	0.079	0.510	0.435	0.469

- **98%** of injected errors found at the right question (born-digital); 98% on scans.
- **3.0** false alarms per clean pair (born-digital).
- **132/132** papers: every question recovered.
- Question labels AP@50: 0.81–0.90.



Errors found per error type

Comparison with Baseline / Existing Methods

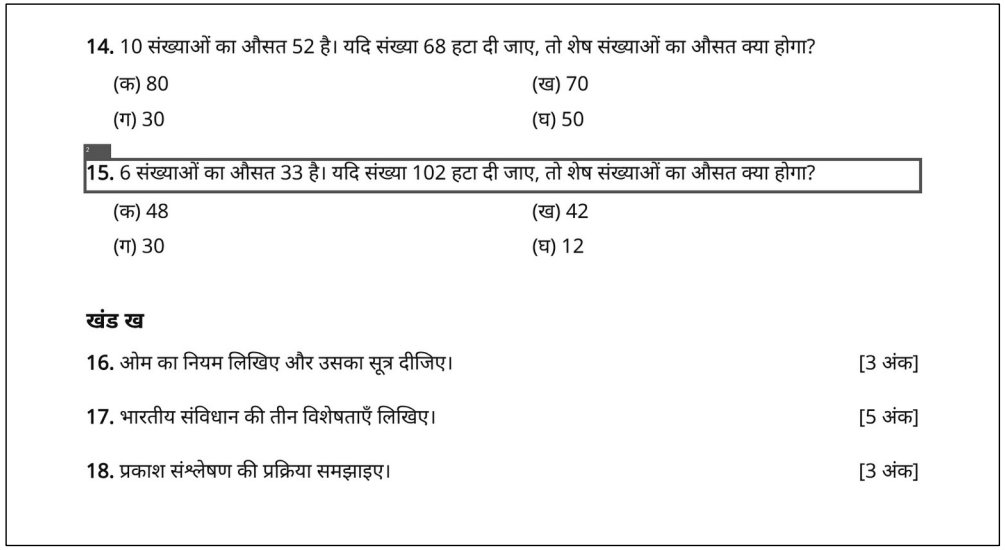
Detector	mAP@50	mAP@50:95
Faster R-CNN (baseline)	0.148	0.079
YOLOv8s (ours)	0.259	0.165
DocLayout-YOLO (used)	0.759	0.575

Verification	Found	False alarms
Full system	98%	4.5
No number pass	77%	13.8
No layout options	98%	11.5
Noisy layout	98%	7.5

Existing approach	Checks paper?	Compares versions?
Manual vetting	by hand	by hand
Answer-sheet AI	no	no
MT quality estimation	no structure	sentences
IndicDoc Verify	**yes**	**question by question**

- Metric: COCO mAP (layout); errors found and false alarms (verification).
- Proposed approach improves: YOLOv8s > baseline; number pass is the largest gain.

Sample Outputs and Demonstration



14. 10 संख्याओं का औसत 52 है। यदि संख्या 68 हटा दी जाए, तो शेष संख्याओं का औसत क्या होगा?

(क) 80 (ख) 70

(ग) 30 (घ) 50

15. 6 संख्याओं का औसत 33 है। यदि संख्या 102 हटा दी जाए, तो शेष संख्याओं का औसत क्या होगा?

(क) 48 (ख) 42

(ग) 30 (घ) 12

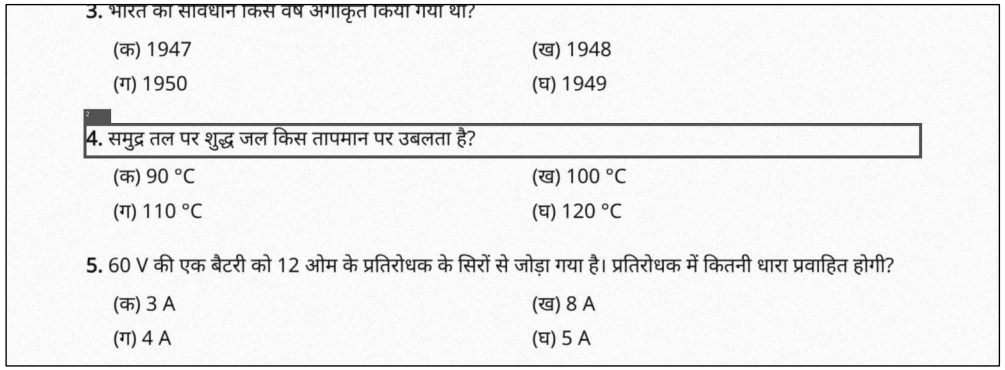
खंड ख

16. ओम का नियम लिखिए और उसका सूत्र दीजिए। [3 अंक]

17. भारतीय संविधान की तीन विशेषताएँ लिखिए। [5 अंक]

18. प्रकाश संश्लेषण की प्रक्रिया समझाइए। [3 अंक]

Correct: *changed number found and outlined*



3. भारत का संविधान किस वर्ष अंगीकृत किया गया था?

(क) 1947 (ख) 1948

(ग) 1950 (घ) 1949

4. समुद्र तल पर शुद्ध जल किस तापमान पर उबलता है?

(क) 90 °C (ख) 100 °C

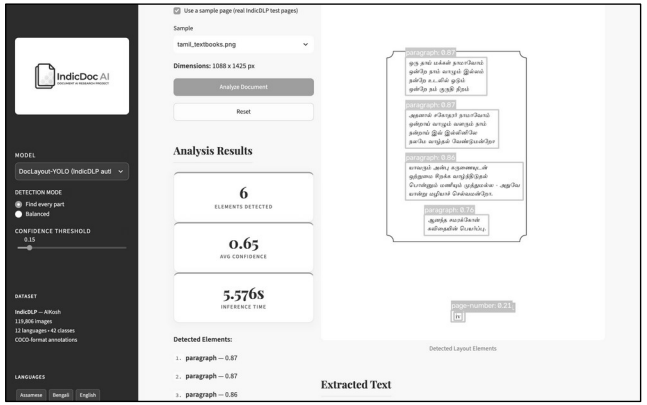
(ग) 110 °C (घ) 120 °C

5. 60 V की एक बैटरी को 12 ओम के प्रतिरोधक के सिरों से जोड़ा गया है। प्रतिरोधक में कितनी धारा प्रवाहित होगी?

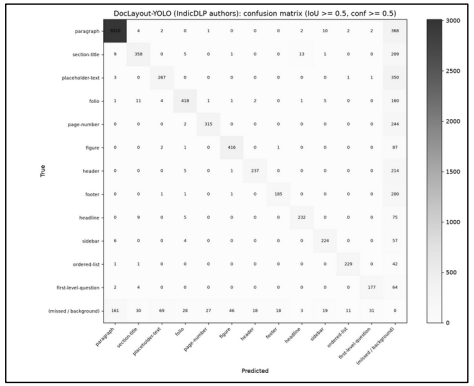
(क) 3 A (ख) 8 A

(ग) 4 A (घ) 5 A

Incorrect: *a false alarm on a degraded scan*



Running app (Tamil page)



Confusion matrix

Challenges and Limitations

- **OCR misreads digits** in Indic text ("1947" → "947"): solved with the number pass.
- **No public data of real errors:** we built a benchmark with injected errors.
- **Meaning is not checked yet:** a dropped word needs a cross-lingual model.
- **Limited compute:** our detectors trained ~1 hour on 2.3K pages, so the product uses the released model.

Conclusion and Future Scope

- **Conclusion**
  - Implemented: layout detection, OCR with number pass, checks, report, app, API, Docker.
  - Results: 98% of errors found; best layout model 0.759 mAP@50.
  - All four objectives met.
- **Future scope**
  - Pilot with an exam body on real papers.
  - Meaning check (cross-lingual model).
  - Better Indic OCR; answer-key check.
  - More document profiles (circulars, forms).

References

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